

Taxonomic identification of small marine organisms is a fundamental yet challenging task in biology and paleontology. Traditional morphological identification relies on expert manual examination, which is subjective, time-consuming, and prone to inconsistencies due to varying levels of expertise. This "taxonomic impediment" severely hampers biodiversity assessment, ecological monitoring, and evolutionary studies. Two groups exemplify this problem: monothalamous foraminifera (single-chambered, soft-shelled protists) and free-living marine nematodes (abundant but morphologically complex metazoans). Both groups are ecologically significant but are often understudied due to identification difficulties, leading to underestimated biodiversity and the proliferation of redundant morphotypes.

Sabbatini et al. (2025) hypothesized that an unsupervised clustering combined with decision trees can objectively group specimens based on simple morphological features, aligning with taxonomic revisions and reducing human bias, even without large image datasets. The objective was to develop a clustering-based method to assist species identification and reduce morphotype redundancy in monothalamous foraminifera (Sabbatini et al., 2025). Brito de Jesus et al. (2023) hypothesized that supervised machine learning algorithms trained on morphometric and categorical character data from the literature can automate species identification with high accuracy, making the process more accessible, efficient, and scalable. The objective was to evaluate the performance of multiple machine learning algorithms in identifying species of two well-known marine nematode genera, *Acantholaimus* and *Sabatieria* (Bruto de Jesus et al., 2023).

Both studies share a core goal: overcoming the taxonomic impediment using machine learning. They address the same fundamental problem: manual identification is slow, subjective, and requires specialized expertise. However, their approaches differ due to data availability and group characteristics. Sabbatini et al. worked with a dataset of heterogeneous, limited images, so they used unsupervised clustering to discover natural groupings and reduce morphotype redundancy. Brito de Jesus et al. worked with groups with rich, well-structured taxonomic literature, enabling them to use supervised classification to assign unknown specimens to predefined species. Both aim to standardize identification and make it accessible to non-experts.

Sabbatini et al. used a dataset of 995 images of monothalamous foraminifera extracted from scientific literature. For each image, 15 morphological features were manually annotated (e.g., basic shape, shell composition, aperture type, etc.). Only 311 specimens were identified to species level; the rest were morphotypes or higher taxa. The model used unsupervised K-Means clustering, with subsequent interpretation and rule extraction using Decision Trees, and compared multiple feature subsets (tests A-F). This was chosen because unsupervised learning allows natural grouping based solely on morphological similarity, which is ideal for detecting morphotype redundancy and exploratory taxonomy (Sabbatini et al., 2025). Brito de Jesus et al. used a dataset composed of morphometric measurements (e.g., body length, amphid diameter) and categorical morphological characters (e.g., head shape, tail shape) compiled from original species descriptions. For *Acantholaimus*: 145 individuals from 40 species; for *Sabatieria*: 260 individuals from 60 species. The model was a supervised classification using four algorithms: Random Forest (RF), Stochastic Gradient Boosting (SGBoos), Support Vector Machine (SVM), and K-Nearest Neighbors (KNN). These were selected to compare a range of established supervised algorithms with different decision boundaries, handling of high-dimensional data, and robustness to overfitting. RF was expected to perform well due to its ability to handle mixed data types and feature interactions.

K-Means is an iterative process that partitions data into k clusters by minimizing the sum of distances between the data points and their central centroid. Decision Tree recursively splits the data based on feature thresholds to maximize class purity at each node, producing interpretable if-then rules. Random Forest builds an ensemble of many decision trees on bootstrapped samples, using random feature subsets at each split, and aggregates votes for classification. SVM maps data into a high-dimensional space and finds the hyperplane that maximizes the margin between classes. SGBost sequentially adds weak decision trees, each correcting the errors of the previous ensemble. KNN classifies a new sample by majority vote among its k nearest neighbors in feature space.

Sabbatini et al.'s model's performance was not evaluated using traditional accuracy metrics because unsupervised clustering lacks ground truth for all specimens. Instead, they assessed cluster homogeneity and alignment with known taxonomy. Test B (7 features) produced 12 clusters, primarily separated by shell nature and basic shape. The Decision Tree from test E showed that basic shape had the highest weight (43%), followed by shell type (26%) and shell nature (21%). The model addresses a critical gap for understudied, soft-shelled taxa using an unsupervised approach that reduces human bias without requiring large labeled datasets. Their decision trees produce interpretable rules aligned with expert taxonomic reasoning. However, this study lacks quantitative performance metrics, relies on subjective manual feature extraction, and uses a small, imbalanced dataset of 995 images. Their model was not deployed as a software tool. The authors propose that the dataset and feature matrix can be provided to researchers, who would manually input morphological observations (from microscopy or images) into the algorithm (Sabbatini et al., 2025).

Brito de Jesus et al.'s model showed that RF achieved the highest accuracy: 93% for *Acantholiamus* (only 1 misclassification out of 14 test individuals) and 100% for *Sabatiera*. SVM also reached 100% for *Sabatiera*, and KNN performed the worst (78% and 93%, respectively). Their model provides a rigorous quantitative evaluation with train/test splitting and cross-validation, achieving up to 100% accuracy and identifying key diagnostic characters via feature importance analysis. Nevertheless, many species (47 of 107) were excluded due to incomplete descriptions, and mean imputation of missing data may introduce bias. The model cannot identify novel species, and morphological variation from different fixation methods may limit generalizability. Their model was deployed via iMESC (Interactive Machine Learning App for Environmental Science), an open-source R-based application. This allows other researchers to load the pre-trained models, run predictions on new data, and retrain with additional data (Brito de Jesus, 2023).

Sabbatini et al. advance the field by demonstrating that unsupervised machine learning can provide objective, reproducible clustering of monothalamids using a small set of manually scored morphological traits, a finding particularly valuable for groups with scarce molecular data and where deep learning is impractical. Their finding that basic chamber shape is the most important feature (43% of the weight) challenges the traditional reliance on cell composition and suggests a revised taxonomic framework for Monothalamia (Sabbatini et al., 2025). In parallel, Brito de Jesus et al. provide strong empirical evidence that supervised machine learning can be achieved using standard morphometric and categorical data, offering a practical, scalable alternative to dichotomous keys with the iMESC deployment lowering the barrier for non-specialists and promoting reproducibility (Brito de Jesus, 2023).

Both papers convincingly show that machine learning—whether supervised or unsupervised— can meaningfully assist marine taxonomic research: Sabbatini et al. is more exploratory and suitable for understudied groups, while Brito de Jesus et al. is more diagnostic and production-ready for well-described groups. Together, they illustrate a continuum of machine learning applications in alpha taxonomy, enabling more accurate species identification across diverse marine organisms.

Citations

- Brito de Jesus, S., Vieira, D., Gheller, P., Cunha, B. P., Gallucci, F., & Fonseca, G. (2023). Machine learning algorithms accurately identify free-living marine nematode species. *PeerJ*, 11, e16216. <https://doi.org/10.7717/peerj.16216>
- Sabbatini, A., et al. (2025). "Machine Learning for identification and classification of Foraminifera: Testing on monothalamids." *Marine Micropaleontology*, 195, 102442. <https://doi.org/10.1016/j.marmicro.2025.102442>